



УНИВЕРСИТЕТ ИТМО

Факультет систем управления и робототехники

Nonlinear systems

Zimenko Konstantin

4. Feedback stabilization

4. Feedback stabilization



What is the task of stabilization with feedback on the state ?

4. Feedback stabilization

✓ State feedback stabilization

For system

$$\dot{x} = f(x, u)$$

the development of designing a feedback control law

$$u = \gamma(x)$$

4. Feedback stabilization

✓ State feedback stabilization

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It is “static
feedback”

4. Feedback stabilization

✓ State feedback stabilization

For system

$$\dot{x} = f(x, u)$$

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$$u = \gamma(x)$$

such that the origin $x = 0$ is asymptotically stable equilibrium point of the closed-loop system

$$\dot{x} = f(x, \gamma(x))$$

4. Feedback stabilization

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such that the origin $x = 0$ is asymptotically stable equilibrium point of the closed-loop system

$$\dot{x} = f(x, \gamma(x))$$

or

$$u = \gamma(x, z)$$

where z is the solution of a dynamical system driven by x ; that is,

$$\dot{z} = g(x, z)$$

4. Feedback stabilization

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It is “dynamic
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where z is the solution of a dynamical system driven by x ; that is,

$$\dot{z} = g(x, z)$$

4. Feedback stabilization

✓ Output feedback stabilization

For system

$$\dot{x} = f(x, u)$$

$$y = h(x, u)$$

the development of designing a feedback control law

$$u = \gamma(y)$$

or

$$u = \gamma(y, z)$$

$$\dot{z} = g(x, z)$$



“static”

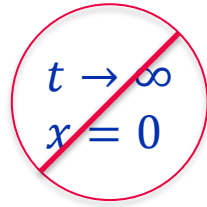


“dynamic”

In the case of dynamic feedback control, the origin to be stabilized is $(x = 0, z = 0)$.

4. Feedback stabilization

✓ What should we do if ...?


$$\begin{array}{l} t \rightarrow \infty \\ x = 0 \end{array}$$



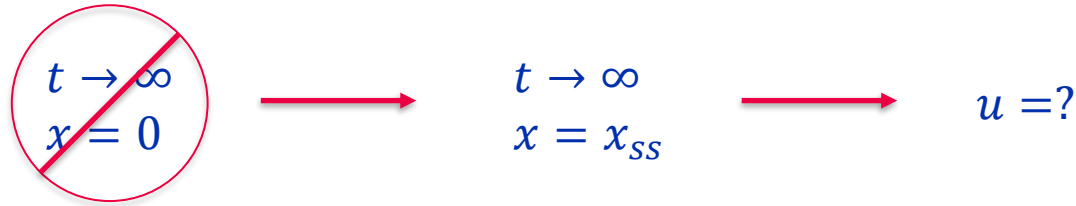
$$\begin{array}{l} t \rightarrow \infty \\ x = x_{ss} \end{array}$$



$$u = ?$$

4. Feedback stabilization

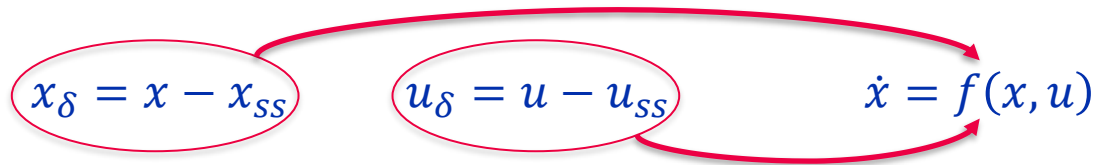
✓ What should we do if ...?



We need to

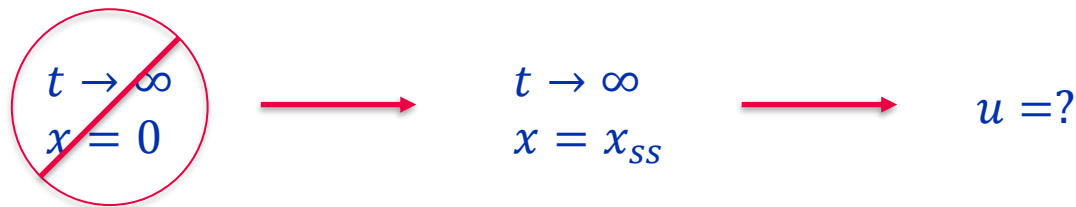
$$0 = \dot{x} = f(x_{ss}, u_{ss}) \quad \forall t \geq 0$$

Replace the coordinates



4. Feedback stabilization

✓ What should we do if ...?



We need to

$$0 = \dot{x} = f(x_{ss}, u_{ss}) \quad \forall t \geq 0$$

Replace the coordinates

$x_{\delta} = x - x_{ss}$ $u_{\delta} = u - u_{ss}$ $\dot{x} = f(x, u)$

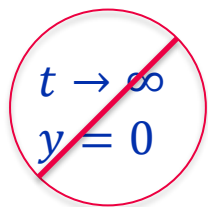
$$\dot{x}_{\delta} + \overset{0}{\cancel{\dot{x}_{ss}}} = f(x_{ss} + x_{\delta}, u_{ss} + u_{\delta}) \stackrel{\text{def}}{=} f_{\delta}(x_{\delta}, u_{\delta})$$

$$f_{\delta}(0, 0) \equiv 0$$

$$u_{\delta} = \gamma(x_{\delta}) \Rightarrow u_{\delta} = u_{ss} + \gamma(x - x_{ss})$$

4. Feedback stabilization

✓ What should we do if ...?

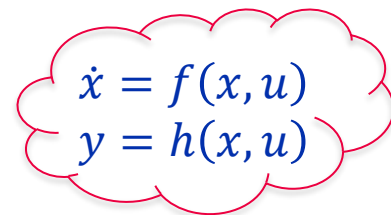

$$\begin{array}{l} t \rightarrow \infty \\ y = 0 \end{array}$$



$$\begin{array}{l} t \rightarrow \infty \\ y = y_{ss} \end{array}$$



$$u = ?$$


$$\begin{array}{l} \dot{x} = f(x, u) \\ y = h(x, u) \end{array}$$

$$\begin{aligned} y_{\delta} &= y - h(x_{ss}, u_{ss}) = \\ &= h(x_{ss} + x_{\delta}, u_{ss} + u_{\delta}) - h(x_{ss}, u_{ss}) \stackrel{\text{def}}{=} h_{\delta}(x_{\delta}, u_{\delta}) \end{aligned}$$

$$\begin{array}{l} \dot{x}_{\delta} = f_{\delta}(x_{\delta}, u_{\delta}) \\ y_{\delta} = h_{\delta}(x_{\delta}, u_{\delta}) \end{array}$$

4. Feedback stabilization

✓ linearization approach

Consider system

$$\dot{x} = f(x, u)$$

Import from linear case

$$\dot{x} = Ax + Bu$$

4. Feedback stabilization

✓ linearization approach

Consider system

$$\dot{x} = f(x, u)$$

Import from linear case

$$\dot{x} = Ax + Bu$$

Assume the pair (A, B) is controllable,
or at least stabilizable

4. Feedback stabilization

✓ linearization approach

Consider system

$$\dot{x} = f(x, u)$$

Import from linear case

$$\dot{x} = Ax + Bu$$

$$A = \left. \frac{\partial f}{\partial x} \right|_{x=0, u=0}$$

$$B = \left. \frac{\partial f}{\partial u} \right|_{x=0, u=0}$$

4. Feedback stabilization

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$$A = \left. \frac{\partial f}{\partial x} \right|_{x=0, u=0} \quad B = \left. \frac{\partial f}{\partial u} \right|_{x=0, u=0}$$

Chose

$$u = -Kx$$

$$\dot{x} = Ax - BKx \longrightarrow \dot{x} = (A - BK)x$$

$(A - BK)$ is Hurwith $\Rightarrow$ the origin an exponentially stable equilibrium point

4. Feedback stabilization

✓ Example: pendulum equation

$$\ddot{\theta} = -a \sin \theta - b \dot{\theta} + cT$$

Stabilize the pendulum at $\theta = \delta$

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$$\ddot{\theta} = -a \sin \theta - b\dot{\theta} + cT$$

Stabilize the pendulum at $\theta = \delta$

$$0 = -a \sin \delta + cT_{ss}$$

$$x_1 = \theta - \delta \quad x_2 = \dot{\theta} \quad u = T - T_{ss} = -Kx$$

4. Feedback stabilization

✓ Example: pendulum equation

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Stabilize the pendulum at $\theta = \delta$

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$$x_1 = \theta - \delta \quad x_2 = \dot{\theta} \quad u = T - T_{ss} = -Kx$$

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -a[\sin(x_1 + \delta) - \sin \delta] - bx_2 + cu \end{aligned} \quad \longrightarrow \quad \dot{x} = Ax + Bu$$

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$$A = \left. \frac{\partial f}{\partial x} \right|_{x=0, u=0} = ?$$

$$B = \left. \frac{\partial f}{\partial u} \right|_{x=0, u=0} = ?$$

4. Feedback stabilization

✓ Example: pendulum equation

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -a[\sin(x_1 + \delta) - \sin \delta] - bx_2 + cu\end{aligned}$$

$$A = \left. \frac{\partial f}{\partial x} \right|_{x=0, u=0} = \begin{bmatrix} 0 & 1 \\ -a \cos(x_1 + \delta) & -b \end{bmatrix}_{x_1=0} = \begin{bmatrix} 0 & 1 \\ -a \cos \delta & -b \end{bmatrix}$$

$$B = \left. \frac{\partial f}{\partial u} \right|_{x=0, u=0} = \begin{bmatrix} 0 \\ c \end{bmatrix} \quad K = [k_1 \quad k_2]$$

4. Feedback stabilization

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$$B = \left. \frac{\partial f}{\partial u} \right|_{x=0, u=0} = \begin{bmatrix} 0 \\ c \end{bmatrix} \quad K = [k_1 \quad k_2]$$

$$A - BK = \begin{bmatrix} 0 & 1 \\ -(a \cos \delta + ck_1) & -(b + ck_2) \end{bmatrix}$$

$$k_1 > -\frac{a \cos \delta}{c} \quad k_2 > -\frac{b}{c}$$

4. Feedback stabilization

✓ Example: pendulum equation

$$\ddot{\theta} = -a \sin \theta - b \dot{\theta} + cT$$

$$0 = -a \sin \delta + cT_{ss}$$

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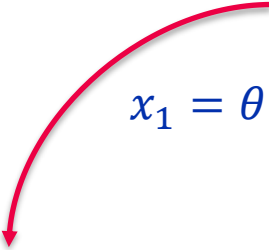
4. Feedback stabilization

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$$0 = -a \sin \delta + cT_{ss}$$

$$x_1 = \theta - \delta \quad x_2 = \dot{\theta} \quad u = T - T_{ss}$$


$$T = T_{ss} + u = \frac{a \sin \delta}{c} - Kx = \frac{a \sin \delta}{c} - k_1(\theta - \delta) - k_2\dot{\theta}$$

$$T = \frac{a \sin \delta}{c} - k_1(\theta - \delta) - k_2\dot{\theta}$$

4. Feedback stabilization

✓ linearization approach (case output)

Consider system

$$\dot{x} = f(x, u)$$

Import from linear case

$$y = h(x)$$

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

$$C = \left. \frac{\partial h}{\partial x} \right|_{x=0, u=0}$$

Assume (A, B) is stabilizable and (A, C) is detectable, and design a linear dynamic output feedback controller

$$\dot{z} = Fz + Gy$$

$$u = Lz + My$$

4. Feedback stabilization

✓ linearization approach (case output)

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$$\begin{aligned}\dot{z} &= Fz + Gy \\ u &= Lz + My\end{aligned}$$

4. Feedback stabilization

$$\begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} Ax + Bu \\ Fz + Gy \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} Ax + B(Lz + My) \\ Fz + GCx \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} Ax + B(Lz + MCx) \\ Fz + GCx \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \underbrace{\begin{bmatrix} A + BMC & BL \\ GC & F \end{bmatrix}}_{\text{Hurwitz closed-loop matrix}} \begin{bmatrix} x \\ z \end{bmatrix}$$

Hurwitz closed-loop matrix

4. Feedback stabilization

✓ Regulation via Integral Control

A red wavy banner with a slight 3D effect and a shadow, containing the word "motivation" in white text.

motivation

4. Feedback stabilization

✓ Regulation via Integral Control

Considered

$$\ddot{\theta} = -\underline{a} \sin \theta - \underline{b} \dot{\theta} + \underline{c} T$$

4. Feedback stabilization

✓ Regulation via Integral Control

Considered

$$\ddot{\theta} = \underbrace{-a \sin \theta}_{\text{System}} \underbrace{- b \dot{\theta}}_{\text{parameters}} + \underbrace{cT}_{\text{parameters}}$$

System
parameters

4. Feedback stabilization

✓ Regulation via Integral Control

Considered

$$\ddot{\theta} = -\underline{a} \sin \theta - \underline{b} \dot{\theta} + \underline{c} T$$

$$T = \frac{a \sin \delta}{c} - k_1(\theta - \delta) - k_2 \dot{\theta}$$

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$$\ddot{\theta} = \underline{-a \sin \theta} - \underline{b\dot{\theta}} + \underline{cT}$$

$$T = \underbrace{\frac{a \sin \delta}{c}}_{T_{ss}} - \underbrace{k_1(\theta - \delta) - k_2\dot{\theta}}_{-Kx}$$

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$$T = \frac{a \sin \delta}{c} - k_1(\theta - \delta) - k_2 \dot{\theta}$$

Let's

$$T_{ss} = \frac{a_0 \sin \delta}{c_0}$$

4. Feedback stabilization

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Let's

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nominal values



4. Feedback stabilization

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$$T = \frac{a \sin \delta}{c} - k_1(\theta - \delta) - k_2 \dot{\theta}$$

Let's

$$T_{ss} = \frac{a_0 \sin \delta}{c_0}$$

$$0 = -a \sin \theta_{ss} - 0 + c \left[\frac{a_0 \sin \delta}{c_0} - k_1(\theta_{ss} - \delta) \right]$$

4. Feedback stabilization

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$$T = \frac{a \sin \delta}{c} - k_1(\theta - \delta) - k_2 \dot{\theta}$$

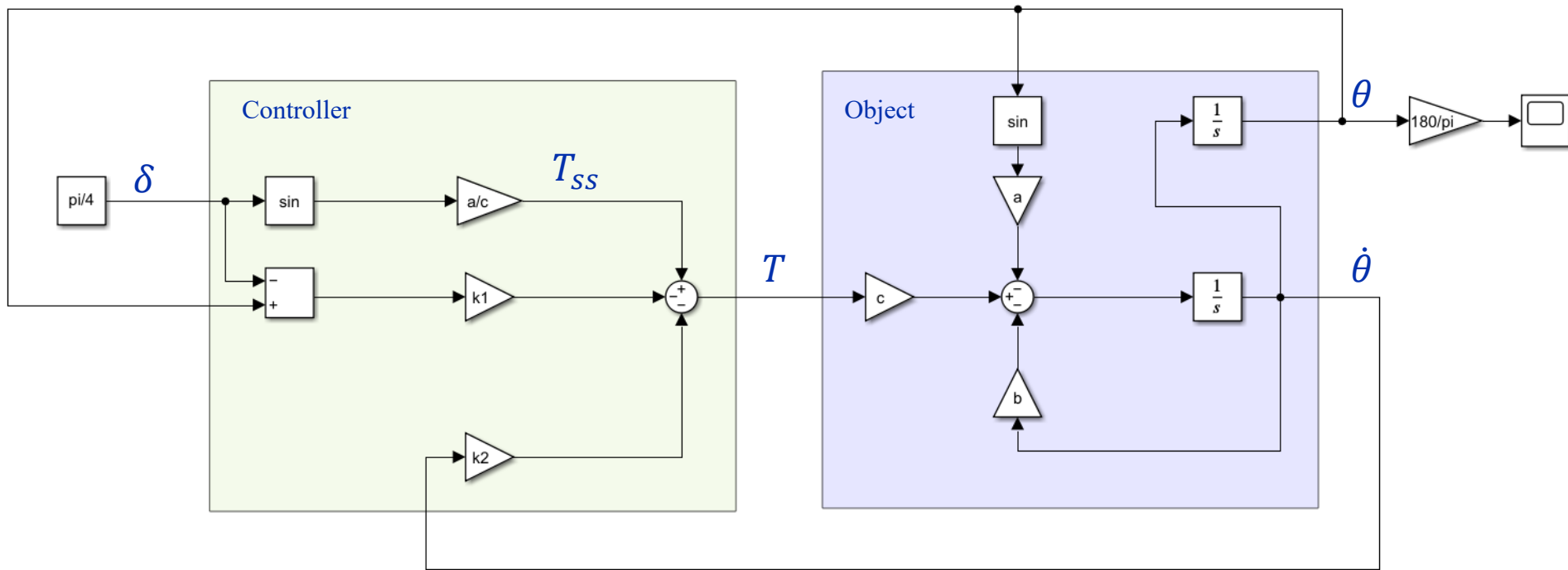
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$$T_{ss} = \frac{a_0 \sin \delta}{c_0}$$

$$a \sin \theta_{ss} = c \left[\frac{a_0 \sin \delta}{c_0} - k_1(\theta_{ss} - \delta) \right]$$

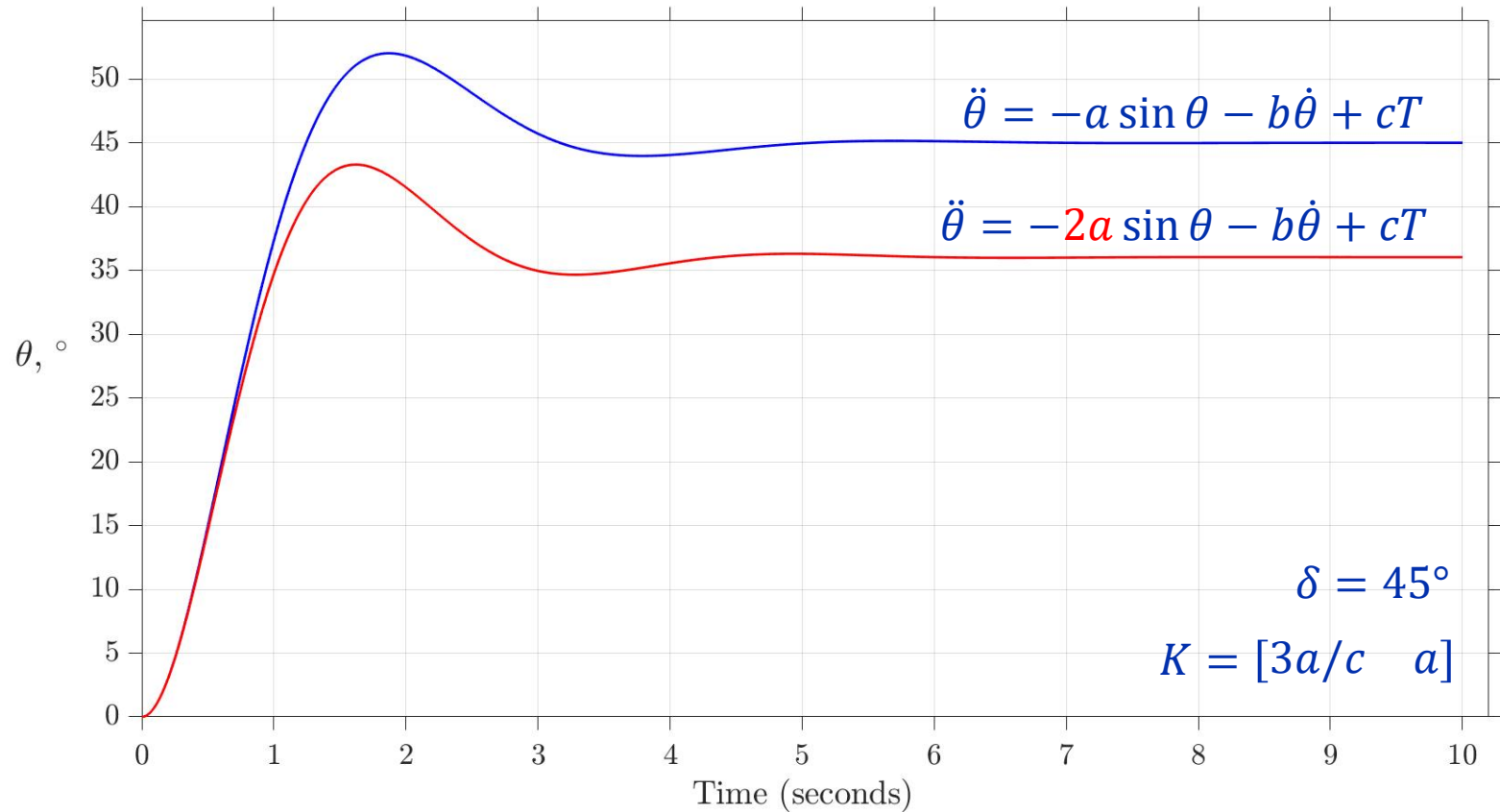
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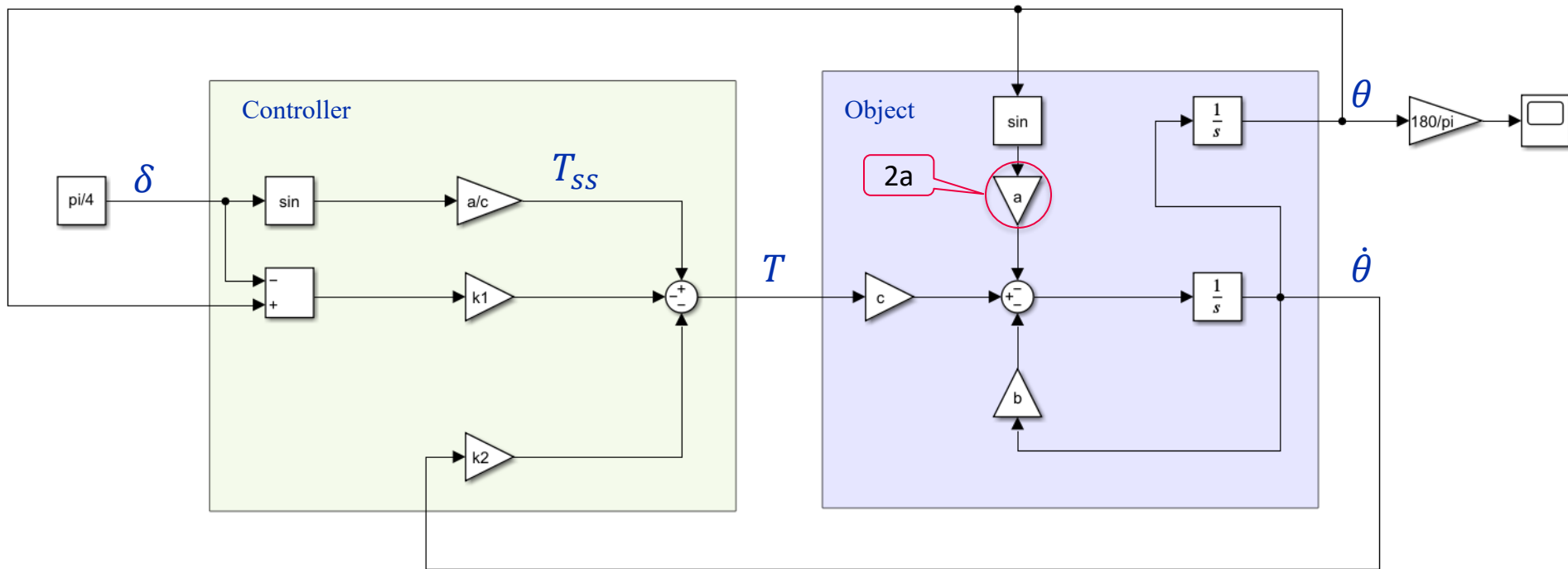
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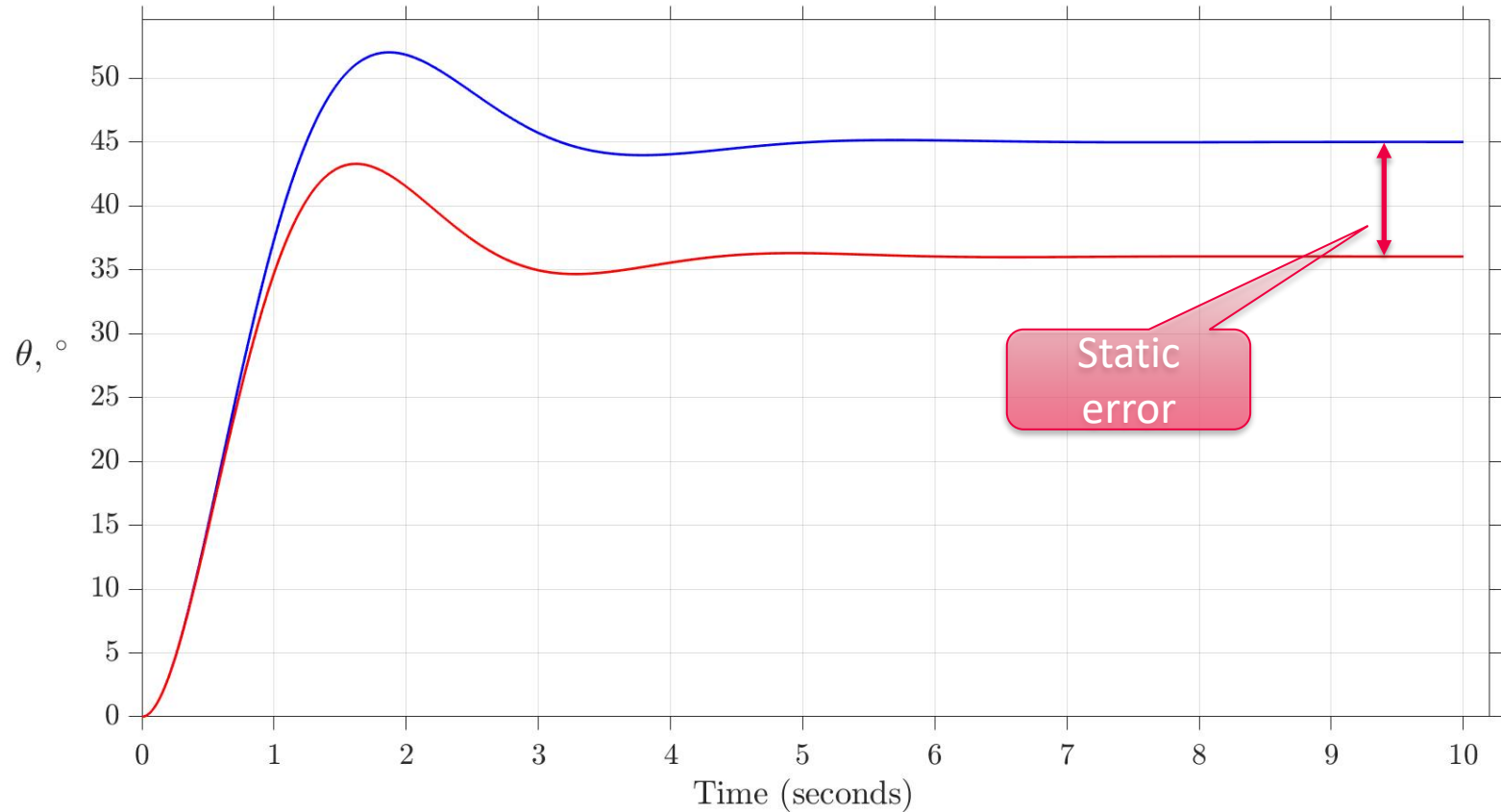
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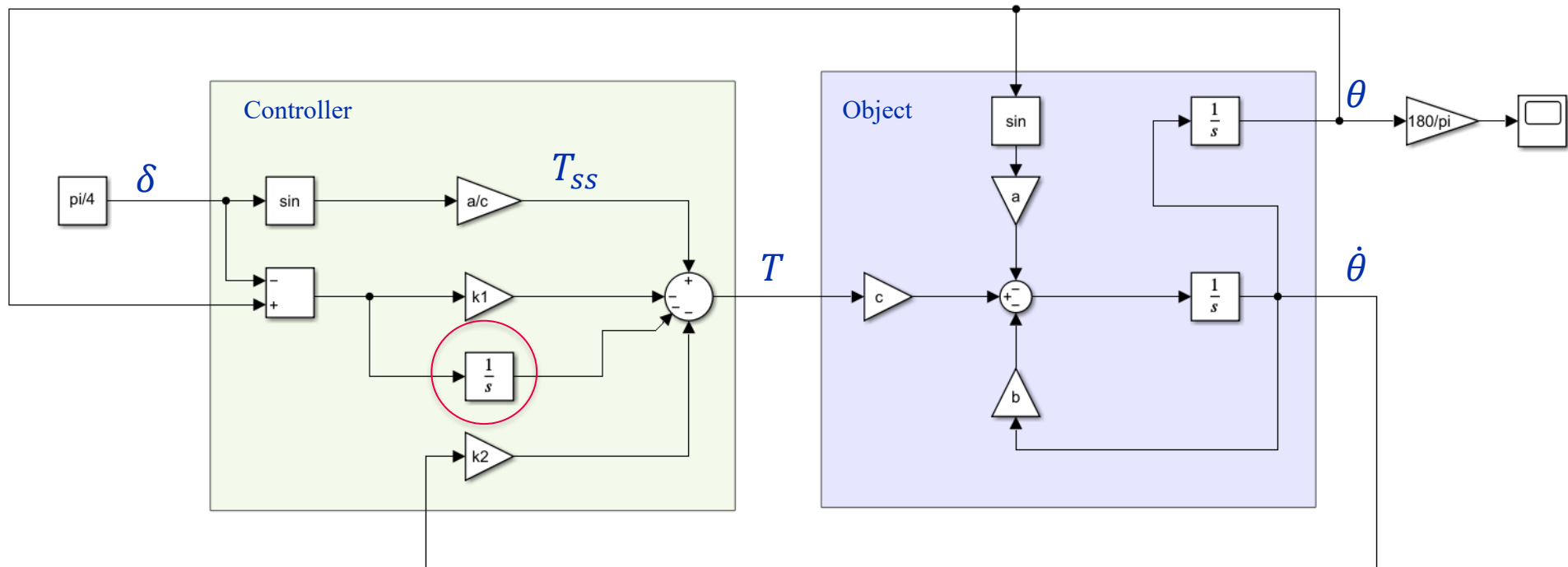
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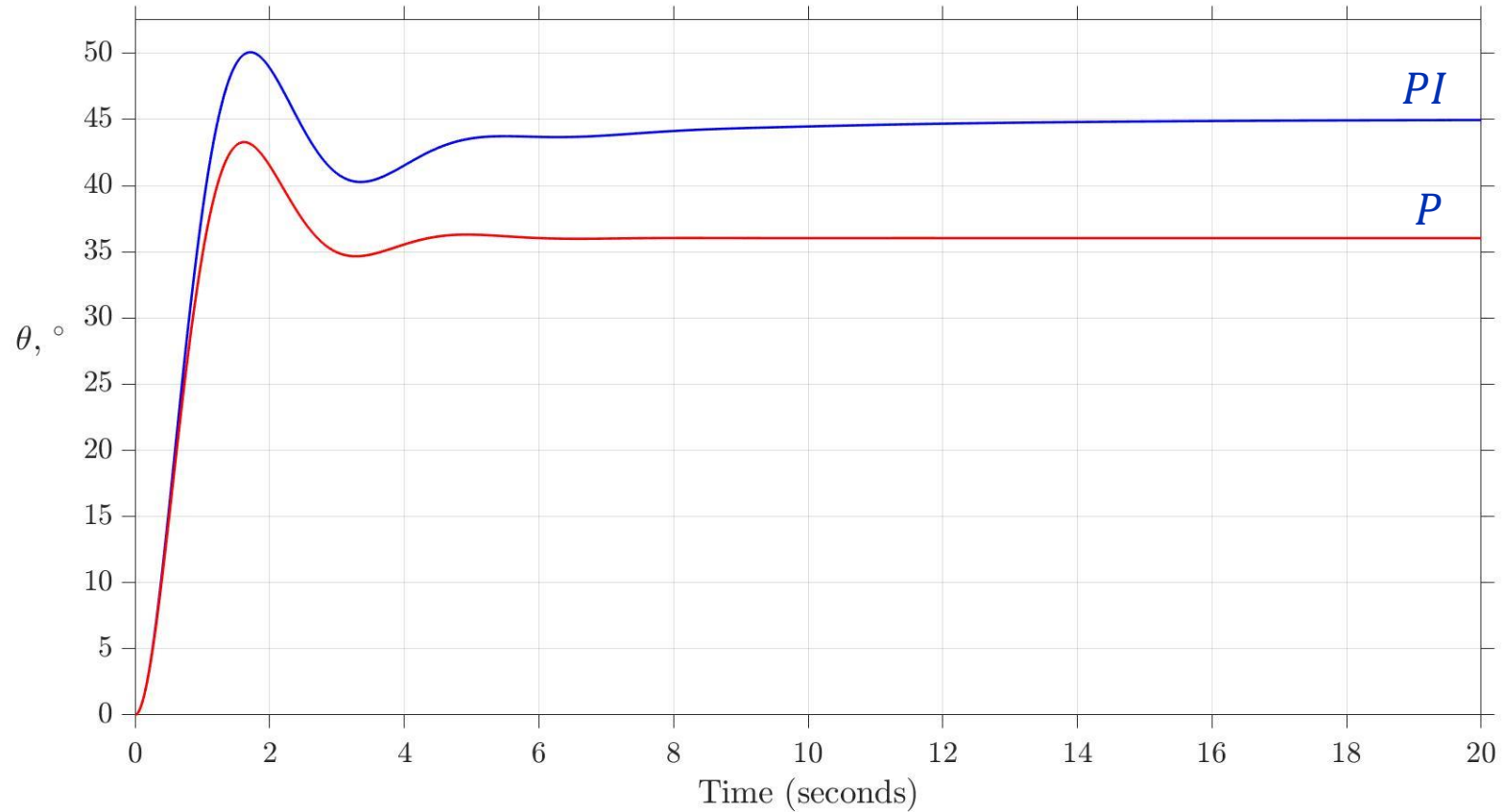
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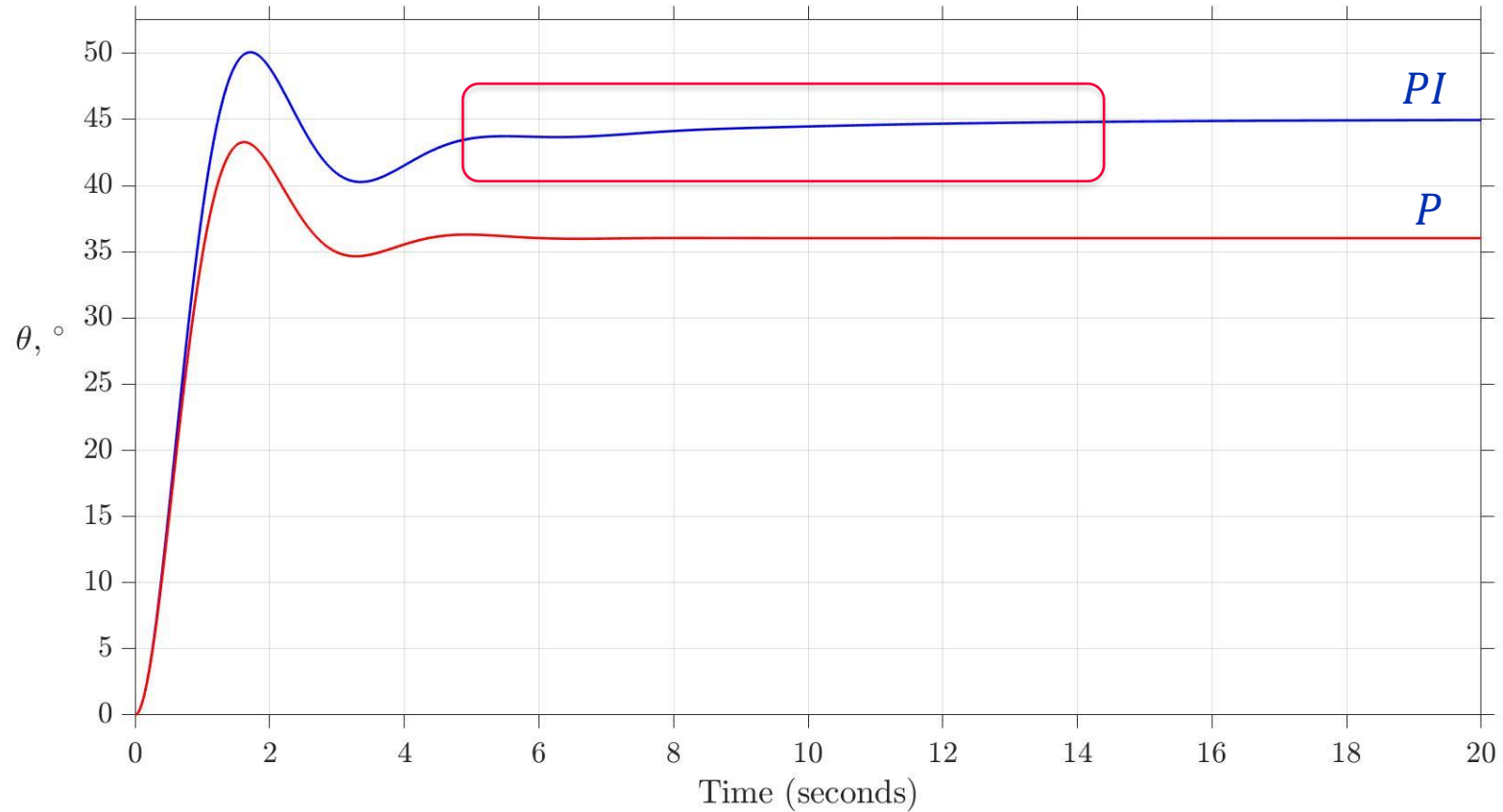
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$$\ddot{\theta} = -a \sin \theta - b\dot{\theta} + cT$$

$$T = \frac{a \sin \delta}{c} - k_1(\theta - \delta) - k_2\dot{\theta} - \int (\theta - \delta) dt$$

$$e = \theta - \delta$$

$$T = \underbrace{\frac{a \sin \delta}{c}}_{T_{ss}} - \underbrace{k_1 e}_{P} - \underbrace{k_2 \dot{\theta} - \int e dt}_I$$

The background is a complex, low-poly geometric pattern in various shades of blue, ranging from dark navy to light sky blue. The shapes are irregular polygons that fit together like a mosaic. In the lower center, there is a white rounded rectangular border containing the text 'IT'sMOre than a UNIVERSITY'.

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UNIVERSITY